

King AbdulAziz University
Faculty of Computing and
Information Technology
Computer Science Department

CPCS – 214
**Computer Organization and
Architecture**

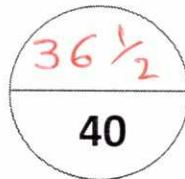
جامعة الملك عبدالعزيز
كلية الحاسبات وتقنية المعلومات
قسم علوم الحاسبات

Exam 2

1st – 1431/1432H

Wednesday 19/ 1 / 1433H

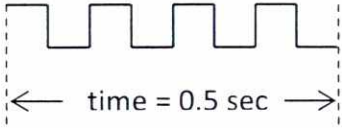
Time: 9:00-10:00 AM



Name: Abdullah Bajaber ID: 0857119 Section: CA

10 1. (10 Points) For each of the following multiple choices choose the correct answer:

✓	1. In a computer system, clock rate is defined as: a. CPU time per instruction b. cycles per second c. cycles per instruction d. duration of a clock cycle
✓	2. Computer performance is improved by: a. increasing CPU time b. increasing clock rate c. increasing response time d. increasing number of clock cycles
✓	3. In a _____, if an element is needed more than once, it must be duplicated. a. R-type instruction b. branch instruction c. single-cycle design d. multi-cycle design
✓	4. In a single-cycle design, clock cycle is determined by the _____ in the machine. a. longest path b. instruction length c. number of instructions d. number of functional units used

5. ✓	_____ are used to alternate data sources used for different instructions. a. ☒ Multiplexors b. Data memory c. Register file d. Instruction memory
6. ✓	In multi-cycle design, when data is used by the same instruction in a later cycle, it must be stored into _____. a. MUX b. state elements c. memory data unit d. ☒ additional registers
7. ✓	The clock rate for the system that uses the clock shown in the next figure is a. 0.125 Hz b. 0.5 Hz c. 4 Hz d. ☒ 8 Hz 
8. ✓	The condition for branch-equal instruction can be tested by performing _____ operation. a. or b. and c. addition d. ☒ subtraction
9. ✓	During execution of a jump instruction, the PC content updated _____. a. 1 time b. ☒ 2 times c. 3 times d. undetermined times
10. ✓	In multi-cycle design, all registers except _____ hold data only between a pair of adjacent clock cycles and no need for write signal. a. ☒ IR b. MDR c. A and B d. ALUOut

2. (10 Points)

A program executes in 10 seconds on computer X, with 1 GHz clock, and CPI of 2.0.

a. Compute performance.

$$2 \quad \text{Performance}_X = \frac{1}{\text{Execution Time}_X} = \frac{1}{10} = 0.1 \text{ s}^{-1}$$

b. Compute CPU clock cycles.

$$\text{Clock cycles}_X = \text{IC} \times \text{CPI} = 5 \times 10^9 \times 2 = 10 \times 10^9 \text{ cycles}$$

OR:

$$\text{Clock cycles}_X = \text{CPU Time}_X \times \text{Clock rate}_X = 10 \times 10^9 \text{ cycles} = 1 \times 10^{10} \text{ cycles}$$

c. Compute Instruction count (IC).

$$\text{CPU Time}_X = \frac{\text{Clock cycles}_X}{\text{Clock rate}_X}$$

$$2 \quad \text{IC} = \frac{\text{CPU Time}_X \times \text{Clock rate}_X}{\text{CPI}} = \frac{10 \times 1 \times 10^9}{2} = 5 \times 10^9 \text{ instructions}$$

d. Design a computer Y with the same clock cycle as computer X, but executes the program in 2 seconds.

$$\text{Clock cycles}_Y = \text{Clock cycles}_X = 10 \times 10^9 = 1 \times 10^{10} \text{ cycles}$$

$$\text{CPU Time}_Y = 2 \text{ s}$$

$$\text{Clock rate}_Y = \frac{\text{Clock cycles}_Y}{\text{CPU Time}_Y} = \frac{1 \times 10^{10}}{2} = 5 \times 10^9 \text{ Hz} = 5 \text{ GHz}$$

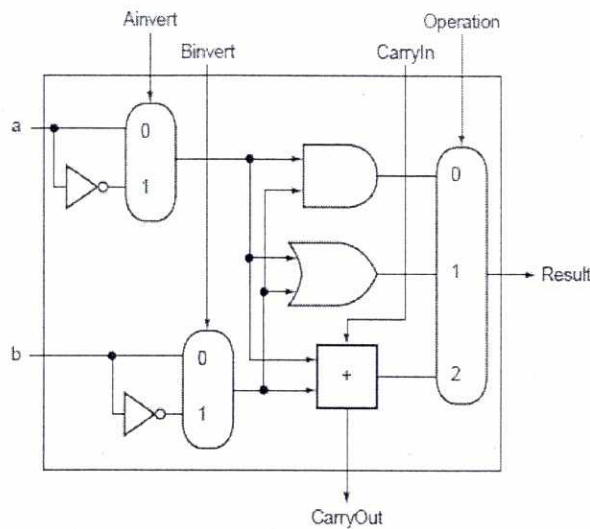
e. Which computer is slower? By how much?

$$\frac{\text{Execution Time}_X}{\text{Execution Time}_Y} = \frac{10}{2} = 5$$

Y is 5 times faster than X

3. (10 Points)

Use the shown digital circuit to answer the following questions.



- a. Explain how to perform **subtract** operation ($b - a$).

$A_{invert} = 1$ ✓
 $B_{invert} = 0$ ✓
 $Operation = 00$ ✓
 Use the AND gate

see the table

a'	b	a	$b - a$ Result
1	0	0	0
1	1	0	1
0	1	1	0
	1	0	?

- 5 b. Using the same above hardware, can you execute the **nand** operation? Justify your answer.

Yes ✓

$A_{invert} = 1$ ✓

$B_{invert} = 1$ ✓

$Operation = 01$ ✓

Using the OR gate we can perform the NAND operation, because:

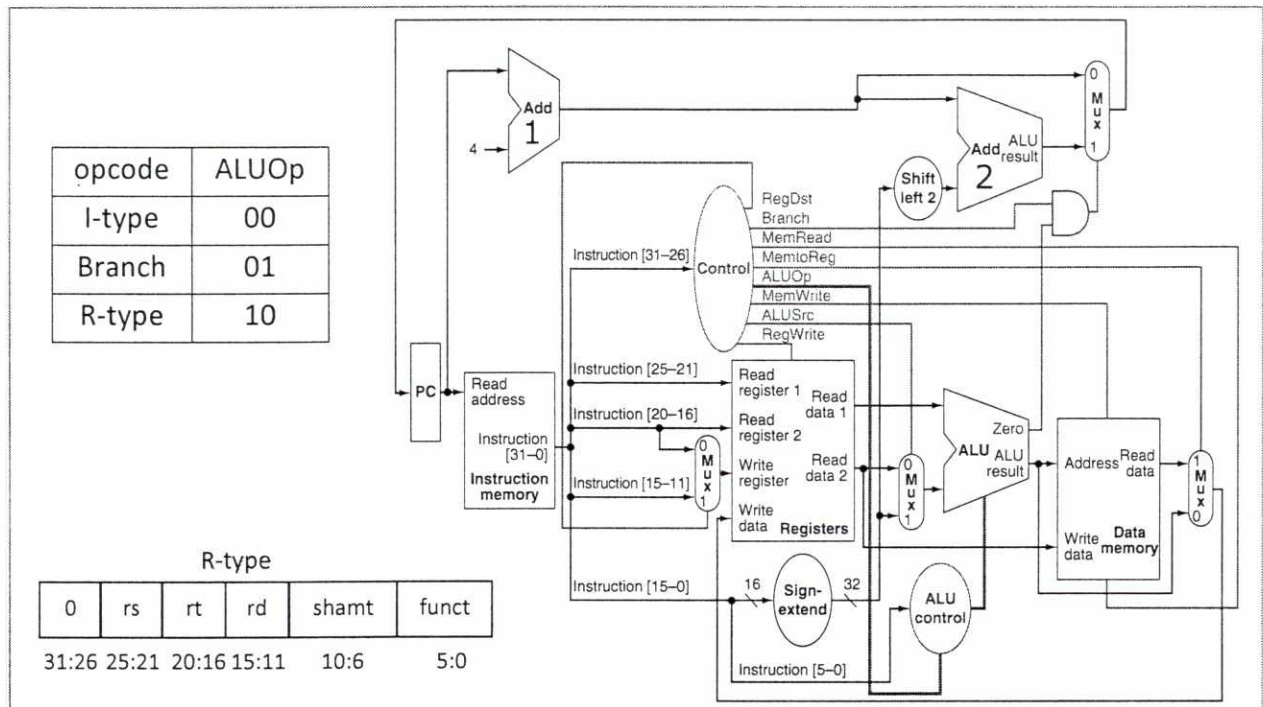
$$(a \text{ AND } b)' \equiv a' \text{ OR } b'$$

a'	b'	a	b	$(a \text{ AND } b)'$
1	1	0	0	1
1	0	0	1	1
0	1	1	0	1
0	0	1	1	0

4. (10 Points)

The instruction: `or $s1, $t3, $t4`

is executed on MIPS datapath processor as shown in the next figure.



- a. Does the design use single-cycle or multi-cycle? Why?
Single-cycle. Because we don't have additional registers to hold the data between multiple cycles.
- b. Check (✓) the functional units used during instruction execution.

Functional Unit	Used	Functional Unit	Used
PC	✓	Instruction memory	✓
Sign extend		Data memory	
Control (Main)	✓	ALU control	✓
Register file	✓	Shift left 2	
ALU	✓	AND gate	
Adder 1	✓	Adder 2	

- c. Write the logic values for setting the various control signals.

Signal	Value	Signal	Value	Signal	Value	Signal	Value
RegDst	1	Branch	0	MemRead	0	ReqWrite	1
MemtoReg	0	MemWrite	0	ALUSrc	0	ALUOp	10

- d. Can you detect **stuck-at-1** fault of the control signal **RegDst** while this instruction is in execution? Justify your answer.

No, I can't detect that, because for this instruction the signal RegDst must be 1.

